

14.1.1 Complex Number Addition using Class

ALGORITHM

1. START

2. DEFINE a class Complex:

- **METHOD** initComplex(self):

- **READ** two space-separated integers (real and imaginary parts) and **STORE** them in self.real and self.imag.

- **METHOD** sum(self, c1, c2):

- **SET** self.real = c1.real + c2.real.

- **SET** self.imag = c1.imag + c2.imag.

- **METHOD** display(self):

- **IF** self.imag >= 0:

- **PRINT** the complex number in the format: real + imag i.

- **ELSE:**

- **PRINT** the complex number in the format: real - abs(imag) i.

3. MAIN PROGRAM:

- **CREATE** three instances of the Complex class: c1, c2, and c3.

- **CALL** `c1.initComplex()` to read the first complex number.

14.1.1. Complex Number Addition using Class

Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- initComplex()**: Read user input for the real and imaginary parts.
- sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- display()**: Display the complex number in the format "**a** + **b**i" or "**a** - **b**i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

Input Format:

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Output Format:

- Print a single line showing the sum of the two complex numbers in the format "**a** + **b**i" or "**a** - **b**i".

Sample Test Cases

```

1 class Complex:
2     def __init__(self):
3         self.real = 0
4         self.imag = 0
5     def initComplex(self):
6         self.real, self.imag = map(int, input().split())
7
8     def sum(self, c1, c2):
9         self.real = c1.real + c2.real
10        self.imag = c1.imag + c2.imag
11
12    def display(self):
13        if self.imag >= 0:

```

Average time: 0.013 s, Maximum time: 0.029 s

4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case	Expected output	Actual output
Test case 1	3.4 -2.5 5 + 9i	3.4 -2.5 5 + 9i
Test case 2		

- **CALL** `c2.initComplex()` to read the second complex number.

- **CALL** `c3.sum(c1, c2)` to compute the addition.
- **CALL** `c3.display()` to output the result.

4. END

